PROPRIETARY & CONFIDENTIAL — LWANDILE ENGINEERING PROJECTS & INCLUSIVE ENTREPRENEURIAL ACADEMY

SCAPUSH PRECISION STANDARD

SECTION 00 — EXPRESSION OF INTEREST (EOI) SUBMISSION

JET Funding Platform — Window 4: Sovereign Food & Energy Ecosystem

Ref: JET-W4-EOI-2026 | Location: Mpumalanga, South Africa | Standard: Scapush Precision | Execution Horizon: 12-Month MVP → 36-Month 2.5MW Scale

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1. Document Header & Governance Information

LISTEN SECTION

Governance Parameter	Reference Details
Lead Execution Entity	Lwandile Engineering Projects (Pty) Ltd (SMME / Technical Delivery Lead)
Governance & Skills Partner	Inclusive Entrepreneurial Academy (Dr. Palesa Charlotte Felix-Faure, Founder)
Principal / Project Lead	Sipho Wiseman Khuzwayo (Inclusive Legacy Architect / Project Lead)
Academic & Governance Lead	Dr. Palesa Charlotte Felix-Faure (Academic Lead & Skills Transfer Principal)
Target Region	Mpumalanga, South Africa
Submission & Execution Baseline	October 2026 Submission | R3.73M MVP Launch → 36-Month 2.5MW Agrivoltaics Expansion
Execution Standard	Scapush Precision Execution Standard
Funding Window	JET Funding Platform — Window 4: Agriculture & Food Systems

2. Executive Summary

LISTEN SECTION

Lwandile Engineering Projects, in strategic partnership with the **Inclusive Entrepreneurial Academy** (founded by Dr. Palesa Charlotte Felix-Faure), formally submits this Expression of Interest (EOI) under Window 4 of the Just Energy Transition (JET) Funding Platform. Recognizing that JET funding strictly evaluates registered, high-capacity institutional partnerships rather than standalone entities, our consortium delivers an end-to-end framework combining rigorous engineering execution with accredited institutional governance and long-term skills transfer.

As coal-fired power infrastructure transitions out of service across Mpumalanga, vulnerable community ecosystems face critical economic challenges. Our consortium deploys an integrated bio-solar and circular agrifood model combining agrivoltaics and controlled-environment hydroponics with an organic waste-to-value biogas and composting pipeline. The immediate rollout is anchored by a **R3.73M Scaled MVP Scope** secured inside a **20,000m² Perimeter Security Fence**.

The operational roadmap is engineered across a **12-Month Buffered MVP Execution Schedule (5,640 Labor Man-Hours & 530 Plant-Hours)** culminating in a **36-Month Expansion Horizon** that scales the campus into a full **2.5MW Agrivoltaics Clean**

Energy & Agrifood Hub. Executed under the Scapush Precision Execution Standard, this alliance guarantees institutional integrity, SARS compliance, and scalable socioeconomic impact.

30kWp → 2.5MW	20,000 m²	12 Months	36 Months
AGRIVOLTAICS SCALING	PERIMETER SECURITY FENCE	BUFFERED MVP COMMISSIONING	2.5MW CAMPUS HORIZON

3. Applicant Eligibility & Institutional Governance Matrix

LISTEN SECTION

Compliance & Partner Parameter	Institutional Role / Status	Details & Strategic Value
Lead Execution Entity	Lwandile Engineering Projects (Pty) Ltd	100% Black-owned South African private enterprise; technical engineering and infrastructure deployment lead.
Governance & Skills Partner	Inclusive Entrepreneurial Academy	Founded by Dr. Palesa Charlotte Felix-Faure; drives institutional governance, curriculum accreditation, and social impact tracking.
Tax & Legal Compliance	Fully Compliant / SARS Good Standing	Registered South African entities in full regulatory compliance with active SARS tax PINs.
Biosecurity & Site Security	20,000m² Fenced Compound	Immediate Month 1 lockdown with a 2.2m heavy-duty perimeter fence securing all bio-solar assets.
JET Alignment	Institutional Strategy Fit	Substantial multi-stakeholder partnership structured for direct grant readiness and governance compliance.

4. Key Focus Areas & Technical Scope

LISTEN SECTION

4.1 Climate-Smart Agriculture & Agrivoltaics Scaling (30 kWp → 2.5 MW)

- **Agrivoltaic Dual-Use Infrastructure:** Immediate deployment of 30 kWp solar PV with 60 kWh LFP BESS, serving as the foundational module for the 36-month expansion into a 2.5MW Agrivoltaics solar canopy providing shading and microclimate control.
- **Sovereign Bio-Solar Barns:** Barn 1 (Poultry - 4,800 broilers) and Barn 2 (Swine - 160 head) integrated with 2% sloped gravity flumes feeding the anaerobic digester.
- **Controlled-Environment Agriculture:** Zone F Nursery equipped with hydronic underfloor heating loops powered by biogas CHP thermal recovery.

4.2 Circular Bioeconomy & Waste-to-Value (70 m³ Digester & 15 kW CHP)

- **Anaerobic Digestion:** Construction of a 70m³ fixed-dome masonry biodigester converting swine slurry and agricultural waste into continuous biogas.
- **Thermal & Electrical Co-Generation:** 15 kW Biogas CHP generator providing off-grid baseload power and hot water loop heat for the nursery and brooding pads.
- **Organic Fertilizer & Composting:** Nutrient-rich digestate processing for soil rehabilitation across surrounding local outgrower farms.

4.3 Governance, Skills Transfer & Accessibility Hub

- **Inclusive Entrepreneurial Academy:** Accredited training modules led by Dr. Palesa Charlotte Felix-Faure, housed in a dedicated 12m container classroom.
- **Zone M Accessibility Hub:** Step-free universal access paving, 4x wheelchair bays, and a 30kW DC EV charger powered by the microgrid.
- **SCADA Telemetry & Control:** 120-tag PLC control center providing real-time data tracking of power, thermal, and biosecurity metrics.

5. Execution Timelines, Budget & 36-Month Expansion Roadmap

LISTEN SECTION

The consortium's operational plan translates immediate R3.73M MVP deployment into a 36-month institutional growth framework. The 12-month MVP construction incorporates 5,640 labor man-hours, 530 plant-hours, and a 2.5-month weather/supply-chain contingency buffer.

Execution Phase	Timeline Horizon	Core Work Packages & Deliverables	Budget & Scaling Focus
Phase I: Site Security, Civil & Digester	Months 1–3	20,000m² perimeter fence, Borehole-01 drilling, 20kL water tower, 70m³ digester masonry bowl.	MVP Capital Deployment (15%)
Phase II: Bio-Solar Barns & Microgrid	Months 4–7	Barn 1 (4.8k broilers), Barn 2 (160 swine), 30kWp PV, 60kWh BESS, 15kW CHP set placement.	MVP Capital Deployment (50%)
Phase III: SCADA, Academy & Commissioning	Months 8–12	Hydronic nursery, SCADA 120-tag desk, Zone M access hub, wet testing & biosecurity stocking.	MVP Capital Deployment (20%)
Phase IV: 2.5MW Campus Expansion	**Months 13–36**	**Scaling agrivoltaics to 2.5MW, expanding outgrower network, commercial Carmello hot chocolate facility & regional microgrid integration.**	**Scale Expansion Horizon (15%)**
TOTAL PROJECT LIFECYCLE	**36 Months**	**Complete 2.5MW Sovereign Agrivoltaics, Bio-Solar & Academy Ecosystem**	**100% JET Grant Framework**

CONSORTIUM GOVERNANCE & BUFFER SAFEGUARDS

Execution Buffer: Months 11–12 function as built-in wet-commissioning and biosecurity stocking buffers, protecting project deadlines against Highveld rainy season delays.

Capital Governance: Escrow-managed financial controls led by Dr. Palesa Charlotte Felix-Faure ensuring complete alignment with JET PMU standards.

6. Execution Framework & Next Steps

LISTEN SECTION

The project will be governed by a Joint Project Steering Committee (JPSC) co-chaired by Sipho Wiseman Khuzwayo and Dr. Palesa Charlotte Felix-Faure, providing transparent oversight across the 12-month MVP and 36-month expansion phases.

Key Implementation Milestones

- **M01–M03:** Perimeter fencing (20,000m²), Borehole-01, and 70m³ digester masonry build.

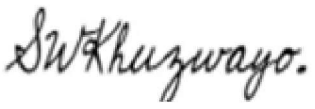
Crucial Submission Detail

- **EOI Submission Target:** November 2026.
- **Execution Standard:** Scapush Precision Execution Standard.

- **M04–M07:** Superstructures for Barns 1 & 2, 30kWp PV array, and 15kW CHP integration.
- **M08–M12:** SCADA online, Zone M accessibility hub, wet commissioning & stocking.
- **M13–M36:** Phased expansion to 2.5MW Agrivoltaics campus and commercial outgrower network.

- **Consortium Leads:** Sipho Wiseman Khuzwayo & Dr. Palesa Charlotte Felix-Faure.

TECHNICAL & PROJECT LEAD SIGNATORY



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Sipho Wiseman Khuzwayo
Inclusive Legacy Architect / Principal Lead
Lwandile Engineering Projects (Pty) Ltd
Status: Digitally Sealed (2026-11-01 23:20:21 UTC) [Scapush Precision]

GOVERNANCE & SKILLS PARTNER SIGNATORY



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Dr. Palesa Charlotte Felix-Faure
Founder & Academic Lead
Inclusive Entrepreneurial Academy
Status: Digitally Sealed (2026-11-03 17:36:16 UTC) [Scapush Precision]

Consortium Leadership
Sipho Wiseman Khuzwayo — Lead System Architect (Lwandile Engineering)
Dr. Palesa Charlotte Felix-Faure — Academic & Governance Lead (Inclusive Entrepreneurial Academy)

Execution Entities & Standard
Lwandile Engineering Projects (Pty) Ltd
Inclusive Entrepreneurial Academy
Scapush Precision Execution Standard

Operational Footprint
Siyabuswa / Witfontein, Mpumalanga Coal Transition Zone, South Africa

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